

# DongGeon (Lloyd) Kwak

Sep 30, 2026 · @lloyd kwak

[kwaklloyd@gmail.com](mailto:kwaklloyd@gmail.com) · [github.com/lloydkwak](https://github.com/lloydkwak) · [linkedin.com/in/lloyd-kwak-347a79435](https://linkedin.com/in/lloyd-kwak-347a79435)

## Education

**Seoul National University of Science and Technology (SeoulTech)** — Seoul, Korea · Feb. 2022 – Present

B.E. in Computer Science and Engineering; Minor in Applied Artificial Intelligence

## Research and Patents

**Undergraduate Researcher**, SeoulTech · Jul. 2025 – Present

Advisor: Prof. Ji-Hyeong Han, Dept. of Computer Science and Engineering. [Research topic: one line to add]

**Patent — Development Contributor** · Korean patent application No. [10-2022-0142398](#) (KIPRIS) · [Patent title to add]

## Work Experience

**Software Developer**, CHASER · Nov. 2022 – Jan. 2023

**Software Developer**, Laver · Jun. 2022 – Oct. 2022

Dates follow the Employment Insurance record (Ministry of Employment and Labor, issued Sept. 30, 2026). [Add one line per role on what you built.]

## Projects

[MakeYourBrick](#) — Single-image to LEGO (LDraw `.ldr`) model generator

- Built a local FastAPI web app that turns one photo into a brick model: SAM2 interactive masks, SAM 3D Objects reconstruction, then a Studio-style color-aware brick tiling.
- Added GLB preview in the browser, automatic base-size selection (16/24/32/48 studs), shadow softening before color quantization, and a mesh-to-LDraw CLI.

- Packaged the GPU runtime in Docker and covered the pipeline with pytest. Python, FastAPI, Docker, WebGL.

### [Frequency-Aware Preprocessing for Diffusion Image Detection](#) — Data-centric AI study

- Detected Midjourney, Stable Diffusion and DALL-E images on the 115,000-image DeepDetect-2025 dataset with a 70:15:15 stratified split.
- Found that pixel statistics differ between real and generated images, and built a frequency-aware preprocessing pipeline around that.
- Raised test accuracy from 72.6% to 96.3% (ResNet-50), 60.7% to 88.9% (EfficientNet-B4) and 78.6% to 92.4% (XceptionNet).

### [Logisim MLP](#) — MNIST digit classifier as a digital circuit

- Implemented a 256-128-10 MLP (about 34,000 weights, ReLU) for 16×16 binarized MNIST digits entirely in Logisim Evolution, with trained weights loaded into ROM.
- Trained the network in Python, mapped the weights into Logisim ROM images, and built the input, hidden and output layers plus a 7-segment result display.

## Certifications and Skills

**Certification** — Craftsman Programming (프로그래밍기능사), HRD Korea · Nov. 2023

**Language** — Korean (Native); English (TOEIC Speaking, Level AL)

**Programming** — Python

**Tools** — PyTorch, FastAPI, Docker, Git and GitHub, OpenCV, pytest